

In Jenkins how to create a folder

There are two common ways to create a folder in Jenkins.

Method 1: Create a Folder from the Jenkins Dashboard (Recommended)

1. Open Jenkins in your browser.
2. Click **New Item** on the left menu.
3. Enter a **Folder Name** (for example: DevOps-Projects).
4. Select **Folder** as the item type.
5. Click **OK**.
6. (Optional) Add a description.
7. Click **Save**.

Now you have a folder named **DevOps-Projects**.

Inside this folder you can create:

- Freestyle Projects
- Pipeline Jobs
- Multibranch Pipeline Jobs
- Other folders

Method 2: Create a Job Inside the Folder

1. Open the folder you created.
2. Click **New Item**.
3. Enter a job name (for example: NodeJS-Deployment).
4. Select **Pipeline**.
5. Click **OK**.
6. Configure your pipeline.
7. Click **Save**.

Your structure will look like this:

Jenkins

|



```
├── DevOps-Projects
│   ├── NodeJS-Deployment
│   ├── React-Deployment
│   └── SpringBoot-Deployment
├── Test
│   └── Docker-Build
└── Production
    └── Deploy-App
```

If you don't see the "Folder" option

The **Folders Plugin** may not be installed.

1. Go to **Manage Jenkins → Plugins**.
2. Search for **Folders**.
3. Install the **Folders Plugin**.

✦ **Upgrade** Jenkins if prompted.



After installation, the **Folder** option will appear when you click **New Item**.

This is the recommended way to organize Jenkins jobs, especially if you have multiple projects such as React, Node.js, Spring Boot, and Docker deployments.

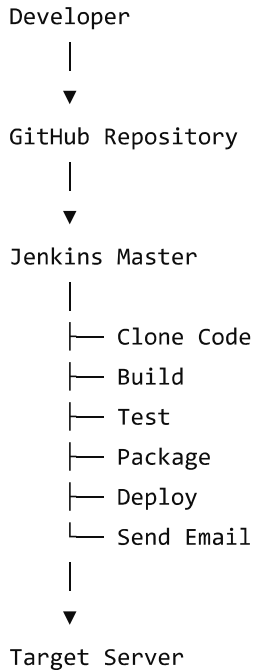


how to job building an application,how to create job and
add agent node or slave
how we can store credentials in jenkins after clone,build
send to email
give me detailed steps



Below is a complete end-to-end Jenkins workflow that is commonly asked in DevOps interviews. It covers creating a job, adding an agent (slave), storing credentials, cloning code, building the application, and sending email notifications.

Jenkins CI/CD Workflow



Step 1: Install Required Software

On Jenkins Server install:

- Java (JDK 17 or 21)
- Git
- Maven (for Java)
- NodeJS (for Node projects)
- Docker (optional)
- Jenkins

Example (Ubuntu)

</> Bash

```
sudo apt update
```

```
sudo apt install openjdk-17-jdk -y
```

```
sudo apt install git -y
```

```
sudo apt install maven -y
```

```
sudo apt install docker.io -y
```

Verify

</> Bash



```
java -version
```

```
git --version
```

```
mvn -version
```

```
docker --version
```

Step 2: Install Jenkins Plugins

Go to

Manage Jenkins



Plugins



Install

- Git Plugin
- Pipeline Plugin
- Maven Integration Plugin
- NodeJS Plugin
- Docker Plugin
- Docker Pipeline
- SSH Agent Plugin
- Email Extension Plugin
- Credentials Plugin

Restart Jenkins.

Step 3: Configure Global Tools

Go to

Manage Jenkins

↓

Tools

Configure

Git

Name

Git

Path

/usr/bin/git

JDK

Name

JDK17

JAVA_HOME

/usr/lib/jvm/java-17-openjdk-amd64

Maven

Name

Maven3

Install Automatically

✓

NodeJS

Name



Node18

Install automatically.

Save.

Step 4: Store GitHub Credentials

Go to

Manage Jenkins



↓

Credentials

↓

System

↓

Global Credentials

↓

Add Credentials

Choose

Kind



Username with password

Fill

Username



your_github_username

Password

GitHub Personal Access Token

ID

github-creds

Description

GitHub Credentials

Save.

Now Jenkins can clone private repositories.

Step 5: Store SSH Credentials (For Deployment)

Again click

Add Credentials



Choose

Kind



SSH Username with private key

Fill

Username



ubuntu

Private Key

Paste Private Key

ID

ec2-ssh

Description

EC2 Server SSH

Save.

Step 6: Configure Email

Go to

Manage Jenkins



System

Find

E-mail Notification



Example

SMTP Server

smtp.gmail.com



Port

587



Enable

Use SMTP Authentication



Username

example@gmail.com



Password

Google App Password



Enable

Use TLS



Test

Test configuration



Save.

Step 7: Add Agent (Slave)

On Agent Machine

Install

</> Bash



```
sudo apt update
```

```
sudo apt install openjdk-17-jdk git -y
```

Create directory

</> Bash



```
mkdir /home/ubuntu/jenkins-agent
```

In Jenkins

Go

Manage Jenkins



↓

Nodes

↓

New Node

Example

Node Name



linux-agent

Select

Permanent Agent



Click OK

Fill

Remote root directory



/home/ubuntu/jenkins-agent

Labels

linux

Usage

Use this node as much as possible

Launch Method

Launch agents via SSH

Host

<Agent-IP>

Credentials

ec2-ssh

Save.

Click

Launch Agent



If successful

Agent is connected



Step 8: Create Pipeline Job

Dashboard



New Item

NodeJS-Build



Choose

Pipeline



OK.

Step 9: Configure Git Repository

Pipeline



Pipeline Script from SCM

SCM

Git



Repository

<https://github.com/username/project.git>



Credentials

github-creds



Branch

main



Step 10: Sample Pipeline

Java Maven Project

groovy



```
pipeline {  
  
    agent { label 'linux' }  
  
    tools {  
        jdk 'JDK17'  
        maven 'Maven3'  
    }  
  
    stages {  
  
        stage('Clone') {  
            steps {  
                git branch: 'main',  
                    credentialsId: 'github-creds',  
                    url: 'https://github.com/username/project.git'  
            }  
        }  
  
        stage('Build') {  
            steps {  
                sh 'mvn clean package'  
            }  
        }  
  
        stage('Test') {  
            steps {  
                sh 'mvn test'  
            }  
        }  
    }  
}
```

```
    }

    stage('Archive') {
        steps {
            archiveArtifacts artifacts: 'target/*.jar'
        }
    }
}

post {

    success {

        mail to: 'yourmail@gmail.com',
        subject: 'Build Successful',
        body: 'Application built successfully.'
    }

    failure {

        mail to: 'yourmail@gmail.com',
        subject: 'Build Failed',
        body: 'Please check Jenkins.'
    }
}

}
```

Step 11: Build Process

Click

Build Now



Stages

Clone

↓

Build

↓



Test



Package



Archive



Email

Step 12: Console Output

Started by user admin



Checking out Git repository

Building project

Running tests

BUILD SUCCESS

Archiving artifacts

Sending email

Finished: SUCCESS

Step 13: Build History

Dashboard



Job



Build History

#1 SUCCESS

#2 SUCCESS

#3 FAILED

Click any build to view:

- Console Output
- Changes
- Test Results
- Artifacts

Step 14: Common Interview Questions

Question	Answer
What is Jenkins?	An automation server used for CI/CD.
What is a Jenkins job?	A task that performs actions like build, test, or deploy.
Difference between Freestyle and Pipeline?	Freestyle is GUI-based; Pipeline is code (Jenkinsfile).
What is an Agent (Slave)?	A machine that executes Jenkins jobs on behalf of the controller.
Why use labels?	To run jobs on specific agents (e.g., <code>linux</code> , <code>docker</code>).
Where are credentials stored?	In Manage Jenkins → Credentials , encrypted and referenced by ID.
How do you clone a private GitHub repo?	Store GitHub credentials and use the <code>credentialsId</code> in the Git configuration.
How do you send email after a build?	Configure SMTP in Jenkins and use the <code>mail</code> step or Email Extension plugin in the <code>post</code> section.

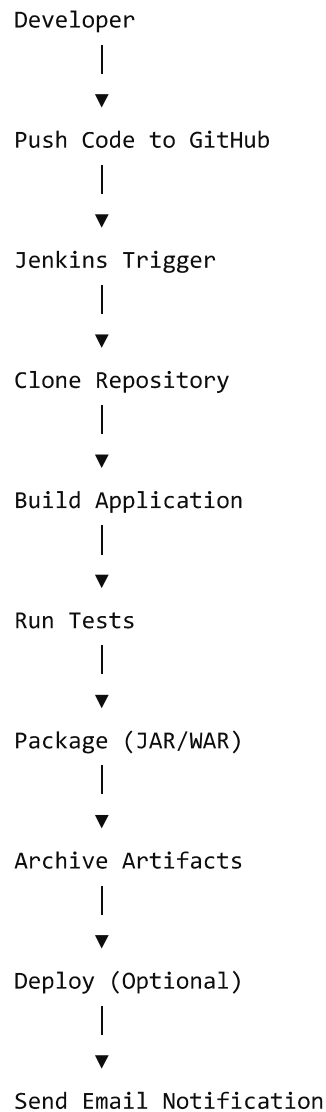
Question

How do you archive artifacts?

Answer

Use `archiveArtifacts` to store build outputs like JARs or WARs.

Complete CI/CD Flow



This workflow represents a typical Jenkins pipeline used in real-world DevOps environments and covers the concepts most commonly discussed in interviews.



Do you like this personality?



ChatGPT can make mistakes. Check important info.